UNITS AND DIMENSIONS

Learning objective: After going through this chapter, students will be able to;

- *understand physical quantities, fundamental and derived;*
- *describe different systems of units;*
- *define dimensions and formulate dimensional formulae;*
- *write dimensionalequations and apply these to verify various formulations.*

1.1 DEFINITION OF PHYSICS AND PHYSICAL QUANTITIES

Physics: Physics is the branch of science, which deals with the study of nature and properties of matter and energy. The subject matter of physics includes heat, light, sound, electricity, magnetism and the structure of atoms.

For designing a law of physics, a scientific method is followed which includes the verifications with experiments. The physics, attempts are made to measure the quantities with the best accuracy.Thus, Physics can also be defined as **science of measurement**.

Applied Physics is the application of the Physics to help human beings and solving their problem, it is usually considered as a bridge or a connection between Physics & Engineering.

Physical Quantities: All quantities in terms of which laws of physics can be expressed and which can be measured are called Physical Quantities.

For example; Distance, Speed, Mass, Force etc.

1.2 UNITS: FUNDAMENTAL AND DERIVED UNITS

Measurement: In our daily life, we need to express and compare the magnitude of different quantities; this can be done only by measuring them.

Measurement is the comparison of an unknown physical quantity with a known fixed physical quantity.

Unit: The known fixed physical quantity is called unit.

OR

The quantity used as standard for measurement is called unit.

For example, when we say that length of the class room is 8 metre. We compare the length of class room with standard quantity of length called metre.

Length of class room = 8 metre

$$\boxed{Q = nu}$$

Physical Quantity = Numerical value × unit

Q = Physical Quantity

n = Numerical value

u = Standard unit

e.g. Mass of stool = 15 kg

Mass = Physical quantity

15 = Numerical value

Kg = Standard unit

Means mass of stool is 15 times of known quantity i.e. Kg.

Characteristics of Standard Unit: A unit selected for measuring a physical quantity should have the following properties

(i) It should be well defined i.e. its concept should be clear.
(ii) It should not change with change in physical conditions like temperature, pressure, stress etc..
(iii) It should be suitable in size; neither too large nor too small.
(iv) It should not change with place or time.
(v) It should be reproducible.
(vi) It should be internationally accepted.

Classification of Units: Units can be classified into two categories.

- **Fundamental**
- **Derived**

Fundamental Quantity: *The quantity which is independent of other physical quantities*. In mechanics, mass, length and time are called fundamental quantities. Units of these fundamental physical quantities are called **Fundamental units.**

e.g. Fundamental Physical Quantity	Fundamental unit
Mass	Kg, Gram, Pound
Length	Metre, Centimetre, Foot
Time	Second

Derived Quantity: *The quantity which is derived from the fundamental quantities e.g. area is a derived quantity*.

Area = Length × Breadth

= Length × Length

= $(\text{Length})^2$

Speed =Distance / Time

=Length / Time

The units for derived quantities are called **Derived Units.**

1.3 SYSTEMS OF UNITS: CGS, FPS, MKS, SI

For measurement of physical quantities, the following systems are commonly used:-

(i) **C.G.S system**: In this system, the unit of length is centimetre, the unit of mass is gram and the unit of time is second.

(ii) **F.P.S system**: In this system, the unit of length is foot, the unit of mass is pound and the unit of time is second.

(iii) **M.K.S**: In this system, the unit of length is metre, unit of mass is kg and the unit of time is second.

(iv) **S.I System**: This system is an improved and extended version of M.K.S system of units. It is called international system of unit.

With the development of science & technology, the three fundamental quantities like mass, length & time were not sufficient as many other quantities like electric current, heat etc. were introduced.

Therefore, more fundamental units in addition to the units of mass, length and time are required.

Thus, MKS system was modified with addition of four other fundamental quantities and two supplementary quantities.

Table of Fundamental Units

Sr. No.	Name of Physical Quantity	Unit	Symbol
1	Length	Metre	m
2	Mass	Kilogram	Kg
3	Time	Second	s
4	Temperature	Kelvin	K
5	Electric Current	Ampere	A
6	Luminous Intensity	Candela	Cd
7	Quantity of Matter	Mole	mol

Table of Supplementary unit

Sr. No	Name of Physical Quantity	Unit	Symbol
1	Plane angle	Radian	rad
2	Solid angle	Steradian	sr

Advantage of S.I. system:

(i) It is coherent system of unit i.e. the derived units of a physical quantities are easily obtained by multiplication or division of fundamental units.

(ii) It is a rational system of units i.e. it uses only one unit for one physical quantity. e.g. It uses Joule (J) as unit for all types of energies (heat, light, mechanical).

(iii) It is metric system of units i.e. it’s multiples & submultiples can be expressed in power of 10.

Definition of Basic and Supplementary Unit of S.I.

1. **Metre** (m): The metre is the length of the path travelled by light in vacuum during a time interval of 1/299 792 458 of a second.

2. **Kilogram** (Kg) : The kilogram is the mass of the platinum-iridium prototype which was approved by the ConférenceGénérale des Poids et Mesures, held in Paris in 1889, and kept by the Bureau International des Poids et Mesures.

3. **Second** (s): The second is the duration of 9192631770 periods of the radiation corresponding to the transition between two hyperfine levels of the ground state of Cesium-133 atom.

4. **Ampere** (A) : The ampere is the intensity of a constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre apart in vacuum, would produce between these conductors a force equal to 2×10^{-7}Newton per metre of length.

5. **Kelvin** (K): Kelvin is the fraction 1/273.16 of the thermodynamic temperature of the triple point of water.

6. **Candela** (Cd): The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian.

7. **Mole** (mol): The mole is the amount of substance of a system which contains as many elementary entities as there are atoms in 0.012 kilogram of Carbon-12.

Supplementary units:

1. ***Radian (rad):*** It is supplementary unit of plane angle. It is the plane angle subtended at the centre of a circle by an arc of the circle equal to the radius of the circle. It is denoted by θ.

   $\theta = l / r$; l is length of the arcand r is radius of the circle
2. ***Steradian (Sr):*** It is supplementary unit of solid angle. It is the angle subtended at the centre of a sphere by a surface area of the sphere having magnitude equal to the square of the radius of the sphere. It is denoted by **Ω**.

$$\Omega = \Delta s / r^2$$

SOME IMPORTANT ABBREVIATIONS

Symbol	Prefix	Multiplier	Symbol	Prefix	Multiplier
D	Deci	10^{-1}	da	deca	10^{1}
c	centi	10^{-2}	h	hecto	10^{2}
m	milli	10^{-3}	k	kilo	10^{3}
µ	micro	10^{-6}	M	mega	10^{6}
n	nano	10^{-9}	G	giga	10^{9}

P f a	Pico femto atto	10^{-12} 10^{-15} 10^{-18}	T P E	tera Pecta exa	10^{12} 10^{15} 10^{18}

Some Important Units of Length:

(i) 1 micron = 10^{-6} m = 10^{-4} cm
(ii) 1 angstrom = 1Å = 10^{-10} m = 10^{-8} cm
(iii) 1 fermi = 1 fm = 10^{-15} m
(iv) 1 Light year = 1 ly = 9.46 x 10^{15}m
(v) 1 Parsec = 1pc = 3.26 light year

Some conversion factor of mass:

1 Kilogram = 2.2046 pound

1 Pound = 453.6 gram

1 kilogram = 1000 gram

1 milligram = 1/1000 gram = 10^{-3} gram

1 centigram = 1/100 gram = 10^{-2} gram

1 decigram = 1/10 gram

1 quintal = 100 kg

1 metric ton = 1000 kilogram

1.4 DEFINITION OF DIMENSIONS

Dimensions: The powers, to which the fundamental units of mass, length and time written as M, L and T are raised, which include their nature and not their magnitude.

For example Area = Length x Breadth

$= [L^1] \times [L^1] = [L^2] = [M^0L^2T^0]$

Power (0,2,0) of fundamental units are called dimensions of area in mass, length and time respectively.

e.g. Density = mass/volume

$= [M]/[L^3]$

$= [M^1L^{-3}T^0]$

1.5 DIMENSIONAL FORMULAE AND SI UNITS OF PHYSICAL QUANTITIES

Dimensional Formula: An expression along with power of mass, length & time which indicates how physical quantity depends upon fundamental physical quantity.

e.g. Speed = Distance/Time

$$= [L^1]/[T^1] = [M^0L^1T^{-1}]$$

It tells us that speed depends upon L & T. It does not depends upon M.

Dimensional Equation: An equation obtained by equating the physical quantity with its dimensional formula is called dimensional equation.

The dimensional equation of area, density & velocity are given as under-

Area = $[M^0L^2T^0]$

Density = $[M^1L^{-3}T^0]$

Velocity = $[M^0L^1T^{-1}]$

Dimensional formula SI& CGS unit of Physical Quantities

Sr. No.	Physical Quantity	Formula	Dimensions	Name of S.I unit
1	Force	Mass × acceleration	$[M^1L^1T^{-2}]$	Newton (N)
2	Work	Force × distance	$[M^1L^2T^{-2}]$	Joule (J)
3	Power	Work / time	$[M^1L^2T^{-3}]$	Watt (W)
4	Energy (all form)	Stored work	$[M^1L^2T^{-2}]$	Joule (J)
5	Pressure, Stress	Force/area	$[M^1L^{-1}T^{-2}]$	Nm^{-2}
6	Momentum	Mass × velocity	$[M^1L^1T^{-1}]$	$Kgms^{-1}$
7	Moment of force	Force × distance	$[M^1L^2T^{-2}]$	Nm
8	Impulse	Force × time	$[M^1L^1T^{-1}]$	Ns
9	Strain	Change in dimension / Original dimension	$[M^0L^0T^0]$	No unit
10	Modulus of elasticity	Stress / Strain	$[M^1L^{-1}T^{-2}]$	Nm^{-2}
11	Surface energy	Energy / Area	$[M^1L^0T^{-2}]$	$Joule/m^2$
12	Surface Tension	Force / Length	$[M^1L^0T^{-2}]$	N/m
13	Co-efficient of viscosity	Force × Distance/ Area × Velocity	$[M^1L^{-1}T^{-1}]$	N/m^2
14	Moment of inertia	Mass × (radius of gyration)^2	$[M^1L^2T^0]$	$Kg\text{-}m^2$
15	Angular Velocity	Angle / time	$[M^0L^0T^{-1}]$	Rad.per sec
16	Frequency	1/Time period	$[M^0L^0T^{-1}]$	Hertz
17	Area	Length × Breadth	$[M^0L^2T^0]$	$Metre^2$
18	Volume	Length × breadth × height	$[M^0L^3T^0]$	$Metre^3$

19	Density	Mass/ volume	$[M^1L^{-3}T^0]$	Kg/m^3
20	Speed or velocity	Distance/ time	$[M^0L^1T^{-1}]$	m/s
21	Acceleration	Velocity/time	$[M^0L^1T^{-2}]$	m/s^2
22	Pressure	Force/area	$[M^1L^{-1}T^{-2}]$	N/m^2

Classification of Physical Quantity: Physical quantity has been classified into following four categories on the basis of dimensional analysis.

1. **Dimensional Constant**: These are the physical quantities which possess dimensions and have constant (fixed) value.

    e.g. Planck's constant, gas constant, universal gravitational constant etc.

2. **Dimensional Variable**: These are the physical quantities which possess dimensions and do not have fixed value.

    e.g. velocity, acceleration, force etc.

3. **DimensionlessConstant**: These are the physical quantities which do not possess dimensions but have constant (fixed) value.

    e.g. e,π $, numbers like$ 1,2,3,4,5 etc.

4. **Dimensionless Variable**: These are the physical quantities which do not possess dimensions and have variable value.

    e.g. angle, strain, specific gravity etc.

Example.1 Derive the dimensional formula of following Quantity & write down their dimensions.

(i) Density (ii) Power

(iii) Co-efficient of viscosity (iv) Angle

Sol. (i) Density = mass/volume

$=[M]/[L^3] = [M^1L^{-3}T^0]$

(ii) Power = Work/Time

=Force x Distance/Time

$=[M^1L^1T^{-2}] \times [L]/[T]$

$=[M^1L^2T^{-3}]$

(iii) Co-efficient of viscosity = $\dfrac{\text{Force x Distance}}{\text{Area x Velocity}}$

$$\frac{\text{Mass x Acceleration x Distance x time}}{\text{length x length x Displacement}}$$

$=[M] \times [LT^{-2}] \times [L]\ [T]/[L^2] \times [L]$

$=[M^1L^{-1}T^{-1}]$

(iv) Angle = arc (length)/radius (length)

= [L]/[L]

$=[M^0L^0T^0]$ = no dimension

Example.2 Explain which of the following pair of physical quantities have the same dimension:

(i) Work &Power (ii) Stress & Pressure (iii) Momentum &Impulse

Sol. (i) Dimension of work = force x distance = $[M^1L^2T^{-2}]$

Dimension of power = work / time = $[M^1L^2T^{-3}]$

Work and Power have not the same dimensions.

(ii) Dimension of stress = force / area = $[M^1L^1T^{-2}]/[L^2] = [M^1L^{-1}T^{-2}]$

Dimension of pressure = force / area = $[M^1L^1T^{-2}]/[L^2] = [M^1L^{-1}T^{-2}]$

Stress and pressure have the same dimension.

(iii) Dimension of momentum = mass x velocity= $[M^1L^1T^{-1}]$

Dimension of impulse = force x time =$[M^1L^1T^{-1}]$

Momentum and impulse have the same dimension.

1.6 PRINCIPLE OF HOMOGENEITY OF DIMENSIONS

It states that ***the dimensions of all the terms on both sides of an equation must be the same***. According to the principle of homogeneity, the comparison, addition & subtraction of all physical quantities is possible only if they are of the same nature i.e., they have the same dimensions.

If the power of M, L and T on two sides of the given equation are same, then the physical equation is correct otherwise not. Therefore, this principle is very helpful to check the correctness of a physical equation**.**

Example: A physical relation must be dimensionally homogeneous, i.e., all the terms on both sides of the equation must have the same dimensions.

In the equation, $S = ut + ½ at^2$

The length (S) has been equated to velocity (u) & time (t), which at first seems to be meaningless, But if this equation is dimensionally homogeneous, i.e., the dimensions of all the terms on both sides are the same, then it has physical meaning.

Now, dimensions of various quantities in the equation are:

Distance, $S = [L^1]$

Velocity, $u = [L^1T^{-1}]$

Time, $t = [T^1]$

Acceleration, $a = [L^1T^{-2}]$

½ is a constant and has no dimensions.

Thus, the dimensions of the term on L.H.S. is $S=[L^1]$ and

Dimensions of terms on R.H.S.

$$ut + \frac{1}{2} at^2 = [L^1T^{-1}]\,[T^1] + [L^1T^{-2}]\,[T^2] = [L^1] + [L^1]$$

Here, the dimensions of all the terms on both sides of the equation are the same. Therefore, the equation is dimensionally homogeneous.

1.7 DIMENSIONAL EQUATIONS, APPLICATIONS OF DIMENSIONAL EQUATIONS;

Dimensional Analysis: A careful examination of the dimensions of various quantities involved in a physical relation is called dimensional analysis. The analysis of the dimensions of a physical quantity is of great help to us in a number of ways as discussed under the uses of dimensional equations.

Uses of dimensional equation: The principle of homogeneity & dimensional analysis has put to the following uses:

(i) Checking the correctness of physical equation.
(ii) To convert a physical quantity from one system of units into another.
(iii) To derive relation among various physical quantities.

1. **To check the correctness of Physical relations**: According to principle of Homogeneity of dimensions a physical relation or equation is correct, if the dimensions of all the terms on both sides of the equation are the same.If the dimensions of even one term differs from those of others, the equation is not correct.

Example 3. Check the correctness of the following formulae by dimensional analysis.

(i) $F = mv^2/r$ (ii) $t = 2\pi\sqrt{l}/g$

Where all the letters have their usual meanings.

Sol. $\boldsymbol{F = mv^2/r}$

Dimensions of the term on L.H.S

Force, $F = [M^1L^1T^{-2}]$

Dimensions of the term on R.H.S

$$\boldsymbol{mv^2/r} = [M^1][L^1T^{-1}]^2 / [L]$$
$$=[M^1L^2T^{-2}]/ [L]$$
$$=[M^1L^1T^{-2}]$$

The dimensions of the term on the L.H.S are equal to the dimensions of the term on R.H.S. Therefore, the relation is correct.

(ii) $\boldsymbol{t = 2\pi\sqrt{l}/g}$

Here, Dimensions of L.H.S, t = $[T^1] = [M^0L^0T^1]$

Dimensions of the terms on R.H.S
Dimensions of (length) = $[L^1]$
Dimensions of g (acc due to gravity) = $[L^1T^{-2}]$

2π being constant have no dimensions.

Hence, the dimensions of terms $2\pi\sqrt{l}/g$ on R.H.S

$$= (L^1/ L^1T^{-2}])^{1/2} = [T^1] = [M^0L^0T^1]$$

Thus, the dimensions of the terms on both sides of the relation are the same i.e., $[M^0L^0T^1]$.Therefore, the relation is correct.

Example 4. Check the correctness of the following equation on the basis of dimensional analysis, $V = \sqrt{\frac{E}{d}}$. Here V is the velocity of sound, E is the elasticity and d is the density of the medium.

Sol. Here, Dimensions of the term on L.H.S

$$V =[M^0L^1T^{-1}]$$

Dimensions of elasticity, E = $[M^1L^{-1}T^{-2}]$
& Dimensions of density, d = $[M^1L^{-3}T^0]$
Therefore, Dimensions of the terms on R.H.S

$$\sqrt{\boldsymbol{\frac{E}{d}}} = [M^1L^{-1}T^{-2}/ M^1L^{-1}T^{-2}]^{1/2} = [M^0L^1T^{-1}]$$

Thus, dimensions on both sides are the same, therefore the equation is correct.

Example 5. Using Principle of Homogeneity of dimensions, check the correctness of equation, h = 2Td /rgCosθ.

Sol. The given formula is, h = 2Td /rgCosθ.
Dimensions of term on L.H.S

Height (h) = $[M^0L^1T^0]$

Dimensions of terms on R.H.S

T= surface tension = $[M^1L^0T^{-2}]$
D= density = $[M^1L^{-3}T^0]$
r =radius = $[M^0L^1T^0]$
g=acc.due to gravity = $[M^0L^1T^{-2}]$
Cosθ = $[M^0L^0T^0]$= no dimensions

So,
Dimensions of 2Td/rgCosθ = $[M^1L^0T^{-2}] \times [M^1L^{-3}T^0] / [M^0L^1T^0] \times [M^0L^1T^{-2}]$

$$= [M^2L^{-5}T^0]$$

Dimensions of terms on L.H.S are not equal to dimensions on R.H.S. Hence, formula is not correct.

Example 6. Check the accuracy of the following relations:

(i) $E = mgh + ½ mv^2$; (ii) $v^3 - u^2 = 2as^2$.

Sol. (i) $E = mgh + ½ mv^2$

Here,dimensions of the term on L.H.S.

Energy, $E = [M^1L^2T^{-2}]$

Dimensions of the terms on R.H.S,

Dimensions of the term, $mgh = [M] \times [LT^{-2}] \times [L] = [M^1L^2T^{-2}]$

Dimensions of the term, $½ mv^2 = [M] \times [LT^{-1}]^2 = [M^1L^2T^{-2}]$

Thus, dimensions of all the terms on both sides of the relation are the same, therefore, the relation is correct.

(ii) The given relation is,

$$v^3 - u^2 = 2as^2$$

Dimensions of the terms on L.H.S

$$v^3 = [M^0] \times [LT^{-1}]^3 = [M^0L^3T^{-3}]$$

$$u^2 = [M^0] \times [LT^{-1}]^2 = [M^0L^2T^{-2}]$$

Dimensions of the terms on R.H.S

$$2as^2 = [M^0] \times [LT^{-2}] \times [L]^2 = [M^0L^3T^{-2}]$$

Substituting the dimensions in the relations, $v^3 - u^2 = 2as^2$

We get, $[M^0L^3T^{-3}] - [M^0L^2T^{-2}] = [M^0L^3T^{-2}]$

The dimensions of all the terms on both sides are not same; therefore, the relation is not correct.

Example 7. The velocity of a particle is given in terms of time t by the equation

$v = At + b/t+c$

What are the dimensions of a, b and c?

Sol. Dimensional formula for L.H.S

$$V = [L^1T^{-1}]$$

In the R.H.S dimensional formula of At

$$[T] = [L^1T^{-1}]$$

$$A = [LT^{-1}] / [T^{-1}] = [L^1T^{-2}]$$

t +c = time, c has dimensions of time and hence is added in t.

Dimensions of t + c is [T]

Now, $b / t + c = v$

$$b = v(t + c) = [LT^{-1}]\,[T] = [L]$$

There dimensions of a= $[L^1T^{-2}]$, Dimensions of b = [L] and that of c = [T]

Example 8. In the gas equation $(P + a/v^2)\,(v - b) = RT$, where T is the absolute temperature, P is pressure and v is volume of gas. What are dimensions of a and b?

Sol. Like quantities are added or subtracted from each other i.e.,

$(P + a/v^2)$ has dimensions of pressure = $[ML^{-1}T^{-2}]$

Hence, a/v^2 will be dimensions of pressure = $[ML^{-1}T^{-2}]$

$$a = [ML^{-1}T^{-2}]\,[\text{volume}]^2 = [ML^{-1}T^{-2}]\,[L^3]^2$$

$$a = [ML^{-1}T^{-2}]\,[L^6] = [ML^5T^{-2}]$$

Dimensions of a = $[ML^5T^{-2}]$

$(v - b)$ have dimensions of volume i.e.,

b will have dimensions of volume i.e., $[L^3]$

or $[M^0L^3T^0]$

2. To convert a physical quantity from one system of units into another.

Physical quantity can be expressed as

$$Q = nu$$

Let n_1u_1 represent the numerical value and unit of a physical quantity in one system and n_2u_2 in the other system.

If for a physical quantity Q;$M_1L_1T_1$be the fundamental unit in one system and $M_2L_2T_2$ be fundamental unit of the other system and dimensions in mass, length and time in each system can be respectively a,b,c.

$$u_1 = [\,M_1^aL_1^bT_1^c]$$

and $$u_2 = [\,M_2^aL_2^bT_2^c]$$

as we know

$$n_1u_1 = n_2u_2$$

$$n_2 = n_1u_1/u_2$$

$$n_2 = n_1 \frac{\left[M_1^a L_1^b T_1^c\right]}{\left[M_2^a L_2^b T_2^c\right]}$$

$$n_2 = n_1 \left[\left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c\right]$$

While applying the above relations the system of unit as first system in which numerical value of physical quantity is given and the other as second system

Thus knowing $[M_1L_1T_1]$, $[M_2L_2T_2]$ a, b, c and n_1, we can calculate n_2.

Example 9. Convert a force of 1 Newton to dyne.

Sol. To convert the force from MKS system to CGS system, we need the equation

$Q=n_1u_1=n_2u_2$

Thus $n_2 = \frac{n_1u_1}{u_2}$

Here $n_1 = 1$, $u_1=1N$, $u_2=dyne$

$$n_2 = n_1 \frac{\left[M_1L_1T_1^{-2}\right]}{\left[M_2L_2T_2^{-2}\right]}$$

$$n_2 = n_1\left(\frac{M_1}{M_2}\right)\left(\frac{L_1}{L_2}\right)\left(\frac{T_1}{T_2}\right)^{-2}$$

$$n_2 = n_1\left(\frac{kg}{gm}\right)\left(\frac{m}{cm}\right)\left(\frac{s}{s}\right)^{-2}$$

$$n_2 = n_1\left(\frac{1000gm}{gm}\right)\left(\frac{100cm}{cm}\right)\left(\frac{s}{s}\right)^{-2}$$

$$n_2 = 1(1000)(100)$$

$$n_2 = 10^5$$

Thus **1N=**10^5 **dynes**.

Example 10. Convert work of 1 erg into Joule.

Sol: Here we need to convert work from CGS system to MKS system

Thus in the equation

$$n_2 = \frac{n_1u_1}{u_2}$$

$n_1=1$

u_1=erg (CGS unit of work)

u_2= joule (SI unit of work)

$$n_2 = \frac{n_1u_1}{u_2}$$

$$n_2 = n_1 \frac{M_1L_1^2T_1^{-2}}{M_2L_2^2T_2^{-2}}$$

$$n_2 = n_1\left(\frac{M_1}{M_2}\right)\left(\frac{L_1}{L_2}\right)^2\left(\frac{T_1}{T_2}\right)^{-2}$$

$$n_2 = n_1\left(\frac{gm}{kg}\right)\left(\frac{cm}{m}\right)^2\left(\frac{s}{s}\right)^{-2}$$

$$n_2 = n_1\left(\frac{gm}{1000gm}\right)\left(\frac{cm}{100cm}\right)^2\left(\frac{s}{s}\right)^{-2}$$

$$n_2 = 1(10^{-3})(10^{-2})^2 \; n_2 = 10^{-7}$$

Thus, **1 erg=** 10^{-7} **Joule**.

Limitations of Dimensional Equation: The method of dimensionshas the following limitations:

1. It does not help us to find the value of dimensionless constants involved in various physical relations. The values, of such constants have to be determined by some experiments or mathematical investigations.
2. This method fails to derive formula of a physical quantity which depends upon more than three factors. Because only three equations are obtained by comparing the powers of M, L and T.
3. It fails to derive relations of quantities involving exponential and trigonometric functions.
4. The method cannot be directly applied to derive relations which contain more than one terms on one side or both sides of the equation, such as v= u + at or s = ut + ½ at^2 etc. However, such relations can be derived indirectly.
5. A dimensionally correct relation may not be true physical relation because the dimensional equality is not sufficient for the correctness of a given physical relation.

* * * * * *

EXERCISES

Multiple Choice Questions

1. [$ML^{-1}T^{-2}$] is the dimensional formula of
   (A) Force
   (B) Coefficient of friction
   (C) Modulus of elasticity
   (D) Energy.

2. 10^5Fermi is equal to
   (A) 1 meter
   (B) 100 micron
   (C) 1 Angstrom
   (D) 1 mm

3. rad / sec is the unit of
   (A) Angular displacement
   (B) Angular velocity
   (C) Angular acceleration
   (D) Angular momentum

4. What is the unit for measuring the amplitude of a sound?

(A) Decibel
(B) Coulomb
(C) Hum
(D) Cycles

5. The displacement of particle moving along x-axis with respect to time is $x=at+bt^2-ct^3$. The dimension of c is
(A) LT^{-2}
(B) T^{-3}
(C) LT^{-3}
(D) T^{-3}

Short Answer Questions

1. Define Physics.
2. What do you mean by physical quantity?
3. Differentiate between fundamental and derived unit.
4. Write full form of the following system of unit
   (i) CGS (ii) FPS (iii) MKS
5. Write definition of Dimensions.
6. What is the suitable unit for measuring distance between sun and earth?
7. Write the dimensional formula of the following physical quantity -
   (i) Momentum (ii) Power (iii) Surface Tension (iv) Strain
8. What is the principle of Homogeneity of Dimensions?
9. Write the S.I & C.G.S units of the following physical quantities-
   (a) Force (b) Work
10. What are the uses of dimensions?

Long Answer Questions

1. Check the correctness of the relation λ = h /mv; where λ is wavelength, h- Planck's constant, m is mass of the particle and v - velocity of the particle.
2. Explain different types of system of units.
3. Check the correctness of the following relation by using method of dimensions
   (i) $v = u + at$
   (ii) $F = mv / r^2$
   (iii) $v^2 – u^2 = 2as$
4. What are the limitations of Dimensional analysis?
5. Convert an acceleration of 100 m/s^2 into km/hr^2.

Answers to multiple choice questions:

1 (C) 2 (C) 3 (B) 4 (A) 5 (C)

ELECTROSTATICS

Learning Objectives : After studying this chapter, the student should be able to;

- *Understand fundamental of charges at rest, properties of point charges;*
- *Explain conservation and quantization of charges;*
- *Relate the properties leading charge storage capacity of the electronic devices using static charges.*

The branch of physics which deals with the study of charges at rest is called electrostatics.

9.1. ELECTRIC CHARGE

Electric Charge: it is the physical property of matter that causes it to experience force when placed in an electromagnetic field. There are two types of charges.

(1) **Positive charge**: e.g. Proton, Alpha particle
(2) **Negative charge**: e.g. Electron, etc.

Charge on electron is smallest unit of charge.

SI unit of charge is coulomb (C).

Charge on Electron = - 1.6×10^{-19} C
Charge on Proton = + 1.6×10^{-19} C

Like charges repel each other and unlike charges attract each other. i.e.

+ ve	+ve	Repel
-ve	-ve	Repel
+ve	-ve	Attract
-ve	+ve	Attract

Conservation of Charge

Charge conservation is the principle that total electric charge in an isolated system never changes. It always remains constant. This also means that no net charge can be created or destroyed. When an atom is ionized, equal amounts of positive and negative charges are produced. Hence the algebraic sum of charges before and after remains the same.

Quantization of Charges

Charge quantization is the principle that the charge of any object is an integer multiple of the elementary charge (e). Thus, an object's charge can be exactly 0 *e*, or exactly 1 *e*, −1 *e*, 2 *e*, etc.,

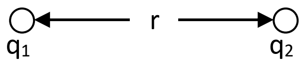
9.2. COULOMB LAW OF ELECTROSTATICS

It states that force of interaction between two point charges is

(i) Directly proportional to magnitude of charges and

(ii) Inversely proportional to the square of the distance between them.

Let F is force between two charges q_1 and q_2. Then



$$F \alpha q_1 q_2$$

$$F \alpha \frac{1}{r^2}$$

$$\Rightarrow F \alpha \frac{q_1 q_2}{r^2} \qquad \text{………(1)}$$

$$F = K \frac{q_1 q_2}{r^2} \qquad \text{………(2)}$$

where K is constant of proportionality and its value is given as

$$K = \frac{1}{4\pi \in_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \text{(in SI units system)}$$

Now from equation (2)

$$F = \frac{q_1 q_2}{4\pi \in_0 r^2} \qquad \text{………..(3)}$$

Here $\in_0$ is electrical permittivity of vacuum. Its value is 8.854×10^{-12} $N^{-1}m^{-2}C^2$.

Let $q_1 = q_2 = q$

$r = 1$ m

then from equation (3) $F = 9 \times 10^9$ N

Thus *one coulomb is that much charge which produces a force of 9 × 10^9 N at a unit charge placed at a distance of 1 m.*

Smaller units of charge; milli Coulomb (mC) = 10^{-3} C.
micro Coulomb (μC) = 10^{-6} C.

9.3. ELECTRIC FIELD

It is the area around the charge in which force of attraction or repulsion can be experienced by another charge.

Electric field intensity at point is defined as the force acting on a unit positive charge at that point.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

- *A unit positive charge is also called as test charge*

The value of q_0 should be very small. Its SI unit is N /C (Newton per Coulomb)

Electric Lines of Force:

An electric line of force is an imaginary continuous line or curve drawn in an electric field such that tangent to it at any point gives the direction of electric force at that point (Fig. 9.1).

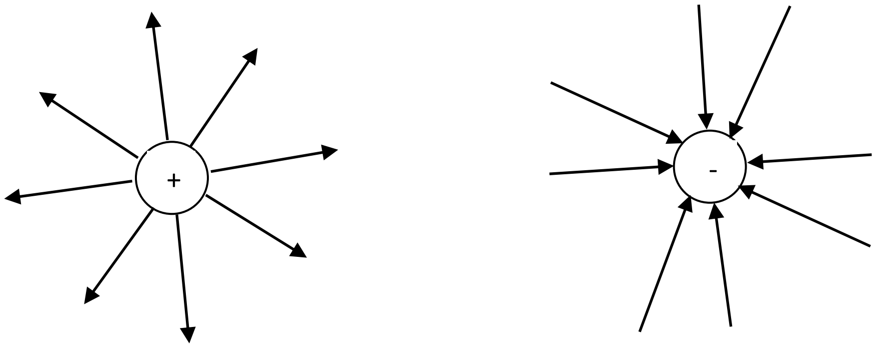


Figure 9.1

Properties of electric lines of force

- *Lines of force originate from a positive charge and terminate to a negative charge.*
- *The tangent to the line of force indicates the direction of the electric field and electric force.*
- *Electric lines of force are always normal to the surface of charged body.*
- *Electric lines of force contract longitudinally and expand laterally.*
- *Two electric lines of force cannot intersect each other.*
- *Two electric lines of force proceeding in the same direction repel each other.*
- *Two electric lines of force proceeding in the opposite direction attract each other. There are no lines of force inside the conductor*

9.4. ELECTRIC FLUX

It is the measure of distribution of electric field through a given surface. Electric flux is defined as total number of electric lines of force passing per unit area normal to the surface. It is denoted by ψ (psi).

Consider small elementary area $\overrightarrow{ds}$ *on a closed surface S. Electric field* $\vec{E}$ *exit in the space. If* θ *is the angle between* $\vec{E}$ *and area vector* $\overrightarrow{ds}$ *as then*

$$\psi = \oint \vec{E}.d\vec{S}$$ *is called electric flux.*

GAUSS'S LAW

It *states that net electric flux of an electric field over a closed surface is equal to the net charge enclosed by the surface divided by* $\in_0$ *i.e.*

$$\psi = \oint \vec{E}.d\vec{S}$$

$$\psi = \oint_S EdS\cos\theta = \frac{q}{\varepsilon_0}$$

Proof: Consider a closed surface S having a charge q placed at a point O inside a closed surface as shown in Fig. 9.2. Take a point P on the surface and consider a small area dS around P.

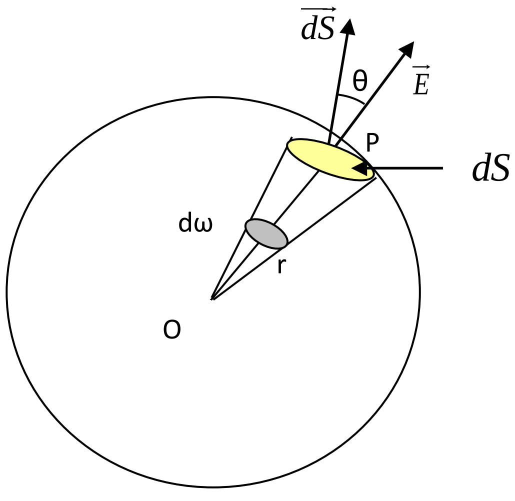


Figure 9.2

Then Electric field at P is

$$E = \frac{q}{4\pi \in_0 r^2} \qquad \ldots\ldots\ldots(1)$$

Now electric flux

$$\psi = \oint_S EdS\cos\theta$$

Putting value of E we get

$$\psi = \oint_S \frac{q}{4\pi\varepsilon_0 r^2} dS\cos\theta$$

$$\psi = \frac{q}{4\pi\varepsilon_0} \oint_S \frac{dS\cos\theta}{r^2}$$

$$\psi = \frac{q}{4\pi\varepsilon_0} \oint_S d\omega \qquad \because \frac{dS\cos\theta}{r^2} = d\omega$$

$$\psi = \frac{q}{4\pi\varepsilon_0}.4\pi \qquad \because Total \quad Solid \quad angle = 4\pi$$

$$\psi = \frac{q}{\in_0}$$

Hence, $\psi = \oint_S EdS\cos\theta = \frac{q}{\varepsilon_0}$

Applications of Gauss's Law:

Electric field due to a point charge:

Consider a point charge q. We want to find electric field at point P at a distance of r from it. Construct a spherical surface of radius r. This is called as Gaussian surface. Consider small area dS on the surface. Let θ is angle between $\vec{E}$ and Area vector as shown in Fig. 9.3.

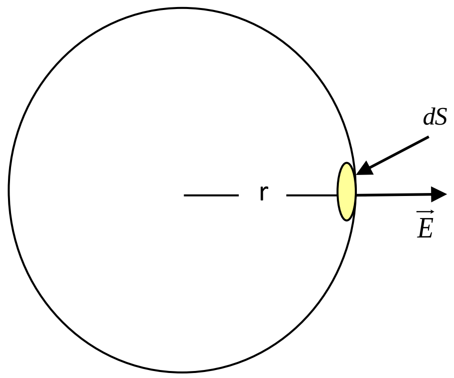


Figure 9.3

Now flux

$$\psi = \oint_S EdS\cos\theta = \frac{q}{\varepsilon_0} \qquad (\because \theta = 0)$$

$$E\oint_S dS = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E.4\pi r^2 = \frac{q}{\in_0} \qquad (\because \text{Area of Sphere} = 4\pi r^2)$$

$$\Rightarrow \boxed{E = \frac{q}{4\pi \in_0 r^2}}$$

Thus the electric intensity decreases with increase in distance.

9.5. CAPACITOR

It is an electronic component that stores electric charge.

Capacitance

The ability of a system to store an electric charge.

As potential is proportional to charge

$$V \propto q$$

or $$q \propto V$$

$$q = CV$$

$$C = \frac{q}{V}$$

Unit of capacitance: Farad (F), microfarad

Grouping of Capacitors

Series Grouping:

A number of capacitors are said be connected in series if -ve plate of one capacitor is connected to the +ve plate of other capacitor and so on.

In this grouping, current is same on each capacitor.

Consider 3 capacitors of capacitances C_1, C_2, C_3 in series. Let V is total applied voltage.

V_1, V_2, V_3 → voltage drops across C_1, C_2, C_3as shown in fig. 9.4.

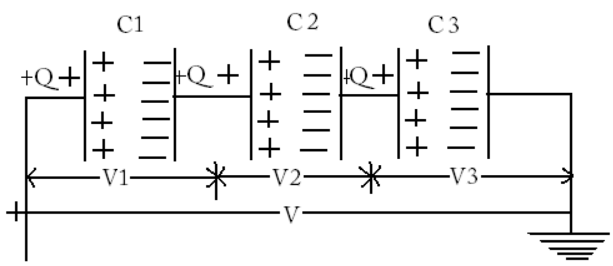


Figure 9.4

Then $V = V_1+V_2+V_3$ ----------------- (1)

Now $C = \frac{q}{V} \Rightarrow V = \frac{q}{C}$

So, $V_1 = \frac{q}{C_1}$, $V_2 = \frac{q}{C_2}$, $V_3 = \frac{q}{C_3}$

Putting in Equation (1)

$$\frac{q}{C} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$$

$$\frac{q}{C} = q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

So the total capacitance decreases in series grouping.

The reciprocal of the equivalent capacitance of two capacitors connected in series is the sum of the reciprocals of the individual capacitances.

Parallel Grouping:

A number of capacitors are said to be connected in parallel if +ve plate of each capacitor is connected to the +ve terminal of battery and –ve plate of each capacitor is connected to the –ve terminal of battery.

In this grouping voltage across each capacitor in same.

Consider 3 capacitors of capacitances C, C_2, C_3 connected in parallel let V is applied voltage.

q_1, q_2, $q_3 \rightarrow$ charges on capacitors C_1, C_2, C_3 as shown in fig. 9.5

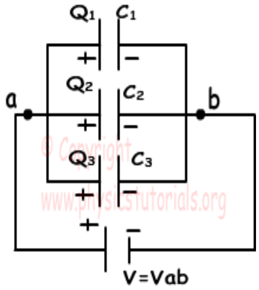


Figure 9.5

So

$$q = q_1 + q_2 + q_3 \text{ ------------------ } (1)$$

Now $C = \dfrac{q}{V}$ or $q = CV$

$\therefore$ $q_1 = C_1 V$, $q_2 = C_2 V$, $q_3 = C_3 V$

Put in equation (1)

$CV = C_1V + C_2V + C_3V$

$CV = (C_1 + C_2 + C_3)\, V$

$C = C_1 + C_2 + C_3$

So the total capacitance increases in parallel grouping.

The equivalent capacitance of capacitors connected in parallel is sum of the individual capacitance.

Solved Numericals

Example 1. Calculate the coulomb force between two protons separated by a distance of 1.6 $\times 10^{-15}$ m.

Solution: Given, 2 protons**;** Charge on Proton= 1.6×10^{-19}C

$q_1 = q_2 = 1.6 \times 10^{-19}$C

Distance, $r = 1.6 \times 10^{-15}$m

Also $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ Nm^2/C^2

Now $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$

$$F = \frac{9\times10^9 \times1.6\times10^{-19} \times1.6\times10^{-19}}{\left(1.6\times10^{-15}\right)^2}$$

F = 90 N

Example 2. Calculate the force between an alpha particle and a proton separated by distance of 5.12×10^{-15}m.

Solution : Given,q_1 = Charge on alpha particle= $2 \times 1.6 \times 10^{-19}$ C

q_2 = Charge on proton=1.6×10^{-19} C

distance, $r = 5.12 \times 10^{-15}$ m

$$\frac{1}{4\pi\varepsilon_o} = 9\times10^9 \text{ Nm}^2/\text{C}^2$$

Now

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$$

$$F = \frac{9\times10^9 \times3.2\times10^{-19} \times1.6\times10^{-19}}{\left(5.12\times10^{-15}\right)^2}$$

F = 17.58 N

Example 3. Three Capacitors of capacitances 3μF, 2μF and 4μF are connected with each calculate total capacitance (a) In Series grouping (b) Parallel grouping.

Solution: Given,

C_1 = 3μF, C_2 = 2μF and C_3 = 9μF

In Series grouping

$$\frac{1}{C_{tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{tot}} = \frac{1}{3} + \frac{1}{2} + \frac{1}{9}$$

$$= \frac{17}{18}\mu F$$

$$\therefore \qquad C_{tot} = \frac{18}{17} = 1.06\mu F$$

In Parallel grouping

$$C_{tot} = C_1 + C_2 + C_3$$

$$C_{tot} = 3 + 2 + 9$$

$$C_{tot} = 14\ \mu F$$

Example 4.Three capacitors 1F, 2F, and 3F are joined in series first and then in parallel. Calculate the ratio of equivalent capacitance in two cases.

Solution: Given,

$C_1 = 1$ F, $C_2 = 2$F, $C_3 = 3$F

In series grouping

$$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_S} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3}$$

$$\frac{1}{C_S} = \frac{11}{6}$$

$$\therefore C = \frac{6}{11} F$$

In Parallel grouping

$$C_p = C_1 + C_2 + C_3$$

$$C_p = 1 + 2 + 3$$

$$C_p = 6F$$

$$\therefore \qquad \text{Ratio} \qquad \frac{Cp}{C_S} = \frac{6}{\frac{6}{11}}$$

$$\text{or} \qquad \frac{Cp}{C_S} = 11$$

EXERCISES

Multiple Choice questions

1) Coulomb's law is only true for point charges whose sizes are
   A. Medium
   B. very large
   C. very small
   D. none of the above
2) As per Coulomb's law, force of attraction or repulsion between two point charges is directly proportional to
   A. sum of the magnitude of charges
   B. square of the distance between them
   C. product of the magnitude of charges
   D. cube of the distance
3) If F is force acting on test charge q_o, electric field intensity E would be given by
   A. $E = F - q_o$
   B. $E = F/q_o$
   C. $E = F + q_o$
   D. $E = q_o/F$
4) In combination of capacitors in series, capacitors are connected
   A. Parallel
   B. side by side
   C. up and down
   D. none of the above
5) Coulomb's force between 2 point charges 10μC and 5μC placed at a distance of 150cm is
   A. 0.2 N
   B. 0.5 N
   C. 2 N
   D. 10 N
6) A device which stores charge is called
   A. Resistor
   B. Inductor
   C. Capacitor
   D. transistor
7) If medium between two charges is air, then value of constant k in SI units will be
   A. $5 \times 10^9\ Nm^2C^{-2}$
   B. $7 \times 10^9\ Nm^2C^{-2}$
   C. $8 \times 10^9\ Nm^2C^{-2}$
   D. $9 \times 10^9\ Nm^2C^{-2}$
8) 1 micro farad (1μF) is equal to
   A. 1×10^{-9} F
   B. 1×10^{-12} F
   C. 1×10^{-6} F
   D. 1×10^{-10} F
9) In equation Q=CV, C is constant of proportionality called capacitor's
   A. Power
   B. Capacitance
   C. Heat

D. Electric intensity

10) The force between two charges is 120 N. If the distances between the charges is doubled, the force will be

A. 60 N
B. 30 N
C. 40 N
D. 15 N

11) SI unit of charge is

A. Coulomb
B. Volt
C. Newton
D. Joule

12) The law governing force between electric charges is

A. Gauss Law
B. Coulomb's law
C. Biot- Savrot Law
D. Ampere's Law

13) Farad is the unit of

A. Capacitance
B. Electric Potential
C. Force
D. Torque

14) Three capacitors of 2μF are joined in parallel;their resultant capacitance is

A. 6μF
B. 10nF
C. 6nF
D. 1.5 μF

15) Unit of Electric intensity is

A. N/Coulomb
B. Joule
C. Newton
D. Coulomb

16) Two capacitors of 2 farad are joined in series their resultant is

A. 1 F
B. 4 F
C. 3 F
D. 0.5 F

Short Answer Questions

1. Define Electric Field
2. What are Electric Lines of force ?
3. Define the termCapacitance.
4. What is Electric flux?
5. Define Capacitor
6. What do you mean by Electric Potential?
7. Define Electric Intensity.
8. State Coulomb's Law
9. Derive expression for electric intensity at a point due to point charge.

10. Explain properties of electric lines of force.
11. Calculate total capacitance in series combination.
12. Calculate total capacitance in parallel combination
13. Explain Gauss's law
14. Define Electric Charge and its types.

Long Answer Type Question

1. Calculate total capacitance when capacitors are connected in series and parallel grouping.
2. State and prove Gauss Law.
3. Using Gauss Theorem find electric field due to a point charge.
4. Derive Coulomb's Law.
5. Derive Coulomb's law from Gauss' law.
6. Use Gauss theorem to derive an expression for the electric field at a point due to a thin infinitely long straight line of charge of uniform charge density ?
7. Derive an expression for the electric field at a point due to uniformly charged spherical shell using Gauss' law.

Answers to Multiple Choice Questions

1)	C	2)	C	3)	B	4)	B
5)	A	6)	C	7)	D	8)	C
9)	B	10)	C	11)	A	12)	B
13)	A	14)	A	15)	A	16)	A

CURRENT AND ELECTRICITY

Learning Objectives: After studying this chapter, the learner should be able to;

- *describe electric current and types of current;AC and DC.*
- *define resistance, combination of resistances; series and parallel.*
- *state Ohm's law, Kirchhoff's law and their applications*

10.1 ELECTRIC CURRENT AND ITS UNITS

In a conductor, there are many free electrons. These electrons are in random motion but there is no net motion along the conductor. But if the two ends of a conductor are at different potentials, the charge will start flowing from one end of conductor to the other end. Therefore, the free electrons (charge) which were moving randomly will now move towards positive terminal of the battery and constitute a current. Hence a potential difference is always needed to make charge move from one end of the conductor to the other end of the conductor.

In a conductor the motion of the free electrons give rise to the electric current as shown in Fig. 10.1.

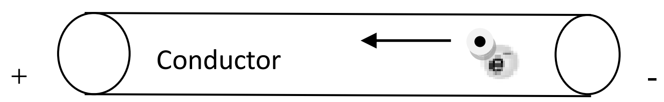


Figure 10.1

Electric current passing through a conductor is *the rate of flow of charge passing through it.* If a charge of q units passes through any cross section of the conductor in t seconds. The current flowing through the wire(I) is given by the formula

$$I = \frac{Charge}{time} = \frac{q}{t}$$

where I = the electric current
q = charge
t = time taken

Unit: Ampere (A)

In the relation

$$I = \frac{q}{t}$$

If the charge is measured in coulombs and time is measured in seconds then the unit of current will be ampere.

Where 1 ampere (A) = $\frac{1\ coulomb}{1\ sec}$

The direction of current is the direction of flow of positive charge i.e. opposite to the direction of flow of electron.

One Ampere:
The current flowing through the conductor is said to be of one ampere if one coulomb of charge flows through the conductor in one second.

Potential Difference (V)

It is the difference in electric potential between two points in an electric circuit, the work that has to be done in transferring unit positive charge from one point to other.

Unit: Volts (V)

One Volt:
Itis defined as energy consumption of one joule per electric charge of one coulomb.

$$1V = \frac{1Joule}{1Coulomb}$$

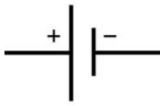
One volt is equal to current of 1 amp times resistance of 1 ohm.

Direct Current

Direct current in an electric wire is that which flow in only one direction. It is the unidirectional flow of current. The electric current flowing through a semi-conductor diode is an example of direct current. Direct current (DC) is produced by sources such as batteries, fuel cells and solar cells and cannot travel over long distances since it has more loss of energy.

The frequency of DC is zero and it has a single polarity. In direct current the electron flow from negative end of the battery to the positive end of the battery.

Symbol of DC voltage source



It can be shown byFig. 10.2.

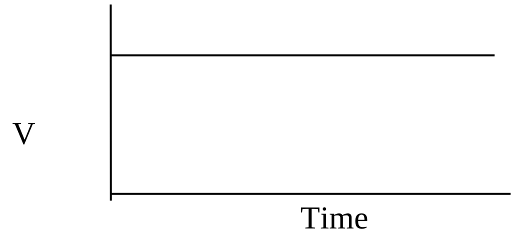


Figure 10.2

DC form is used in low voltage apparatus like charging batteries, cell phones, automotive apparatus, aircraft apparatus and other low voltage low current apparatus.

AC (Alternating current)

As shown in Fig. 10.3,AC is current that reverses the direction periodically and also has a magnitude that varies continuously with time.

AC is used in our homes. Power stations generate ac because it is easy to low and raise the voltage with the help of transformers.In North America the frequency of AC is 60 Hz and in India it is 50 Hz. The AC in our home is sinusoidal in nature.

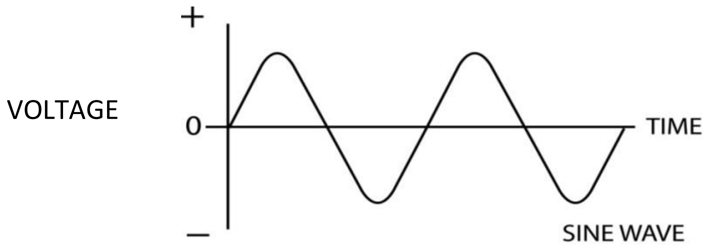


Figure 10.3

The radio frequency current in antennas and transmission lines are the examples of AC.

Symbol of AC

It is produced by an alternator and has more power and can be easily transferred from one place to another.

10.2 OHM'S LAW

According to Ohm's law "*The current flowing through a conductor is always directly proportional to the potential difference between the two ends if the physical condition (temperature, pressure etc.) of the conductor remains the same*".

If I is the current passing through a conductor and and V is the potential difference between the ends of the conductor then

$$V \propto I$$

$$V = R\,I$$

$$\frac{V}{I} = R$$

Therefore, $$R = \frac{V}{I} = \frac{\text{Potential Differnce}}{\text{Electric Current}}$$

where R is a constant and is called electric resistance.

The value of R depends upon nature, dimension and temperature of the conductor.

$$V = I\,R$$

Therefore $$I = \frac{V}{R}$$

If a graph is drawn between current (I) and the potential difference (V) it will be a straight line for a conductor (Fig. 10.4).

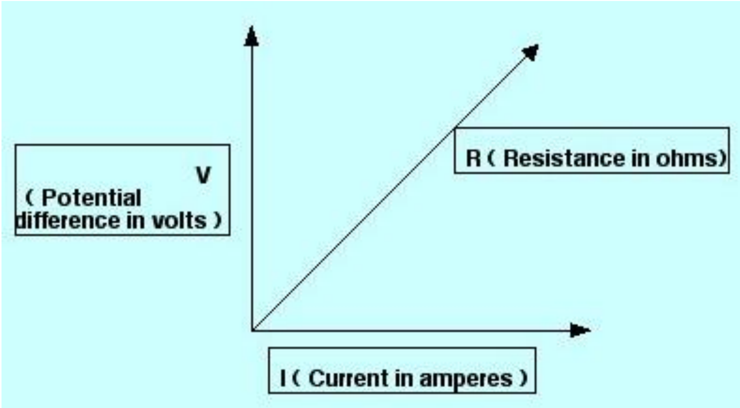


Figure 10.4

10.3 RESISTANCE (R)

The opposition to the flow of electric current in an electric circuit is called resistance. Therefore, it is the measure of the difficulty to pass an electric current through the circuit.

$$R = \frac{V}{I} = \frac{Potential\ difference}{Electric current}$$

If V is measured in volts and I is measured in amperes then the resistance R is measured in ohms.

Symbol:

Unit: Ohms (Ω)

One Ohm:

$$1\ \text{ohm} = \frac{1\ Volt}{1\ Ampere}$$

Therefore, one ohm is the resistance of conductor in which a current of one ampere flows through it when the potential difference of one volt is maintained between its two ends.

Specific Resistance (Definition and Units)

The resistance of a conductor depends on following factors;

(i) The resistance of a given conductor is directly proportional to its length i.e.

$$R \propto l \qquad \text{............ (1)}$$

ii) The resistance of a given conductor is inversely proportional to its area of cross-section.

$$R \propto \frac{1}{A} \qquad \text{..........(2)}$$

By combining equation (1) and (2), we get

$$R \propto \frac{l}{A}$$

or $$R = \rho \frac{l}{A}$$

where ρ (rho) is a constant and known as specific resistance or resistivity of the material. The resistivity of a material does not depend on its length or thickness. It depends on the nature of the material.

If r = *1* m *and A = 1 m*^{2} *then from above equation*

$$\rho = \mathrm{R}$$

Thus resistivity of the material is the resistance of a conductor having unit length and unit area of cross- section.

Units: Ohm-m (Ωm)

Conductivity

It is the degree to which an object conducts electricity. This is the reciprocal of the resistivity,

$$\sigma = \frac{1}{\rho}$$

Where, σ is the conductivity and ρ is the resistivity of the conductor.
Unit: Siemens per meter or mho per meter

Conductance (G)

It is the reciprocal of the resistance and it is a measure of ease with which the current flows through a substance.

$$\mathrm{G} = \frac{1}{R}$$

where G = Conductance
R = Resistance

Unit: mho

10.4 COMBINATION OF RESISTANCES

1. Series combination

The resistance are said to be connected in series if the same current passes through all the resistances and the potential difference is different across each resistance.

Let three resistances R_1, R_2, R_3 be connected in series as shown in the Fig. 10.5

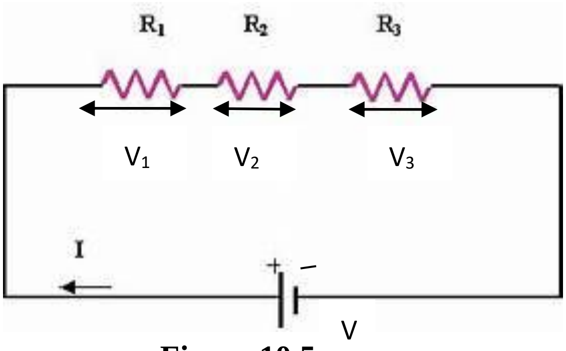


Figure 10.5

Let

V = Voltage applied across the series combination

I = Current passing through the circuit

Clearly current I is same throughout the circuit

Let V_1, V_2, V_3 be the potential difference across R_1, R_2, R_3 respectively. Then, according to Ohm's law

$$V = I R$$

where R is the total resistance in series

Now

$$V = V_1 + V_2 + V_3 \qquad \text{---------------} \ (1)$$

Then by Ohm's law

$$V_1 = I R_1$$
$$V_2 = I R_2$$
$$V_3 = I R_3$$

Putting the values of V_1, V_2 and V_3 in equation (1) we get

$$IR = I R_1 + I R_2 + I R_3$$
$$IR = I (R_1 + R_2 + R_3)$$
$$R = (R_1 + R_2 + R_3)$$

Thus the combined resistances when they are connected in series is the sum total of the individual resistances.

2. Parallel Combination

The resistances are said to be connected in parallel if the potential difference across each resistance is the same but the current passing through each resistance is different.

Let there be three resistances R_1, R_2, R_3 connected in parallel as shown in Fig. 10.6. One end of each resistance is connected to point A and the other end of each resistance is connected to the point B.

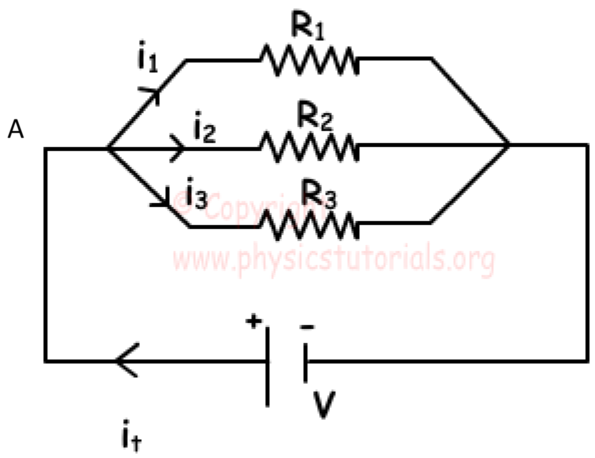


Figure 10.6

Let

V = Potential Difference applied across A and B

Clearly, potential difference V is same across each resistance.

Let I = total current flowing in the circuit.

R = total resistance of the circuit

Let I_1, I_2, I_3 be the current passing through the resistances R_1, R_2, R_3 respectively.

From Ohm's law applied to the whole circuit

$$I_1 = \frac{V}{R_1}$$

$$I_2 = \frac{V}{R_2}$$

$$I_3 = \frac{V}{R_3}$$

Now we have,

$$I = I_1 + I_2 + I_3 \qquad \text{-----------------------} (2)$$

Putting the values of I, I_1, I_2, I_3 in the equation (2)

$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$V\frac{1}{R} = V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$$

Or
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Thus we can say that if the resistances are connected in parallel, then the reciprocal of the equivalent resistance is equal to the sum of reciprocals of individual resistances in the circuit.

10.5 HEATING EFFECT OF ELECTRIC CURRENT

When an electric current is passed through a conductor, the conductor becomes hot after some time and produces heat. This happens due to the conversion of some electric energy passing through the conductor into heat energy. This effect of electric current is called heating effect of current.

The heating effect of current was studied experimentally by Joule in 1941. After doing this experiments, Joule came to the conclusion that the heat produced in a conductor is directly proportional to the product of square of current (I^2), resistance of the conductor (R) and the time (t) for which current is passed. Thus,

$$H = I^2Rt$$

Derivation of Formula

To calculate the heat produced in a conductor, consider current I is flowing through a conductor of resistance R for time t. Also consider that the potential difference applied across its two ends is V.

Now, total amount of work done in moving a charge q from point A to point B is given by:

$$W = q \times V \qquad \text{------------------ (1)}$$

Now, we know that charge = current x time

$$\text{or} \quad q = I \times t$$

$$\text{and} \quad V = I \times R \quad \text{(Ohm's law)}$$

Putting the values of q and V in equation (1), we get

$$W = (I \times t) \times (I \times R)$$

$$\text{or} \quad W = I^2Rt$$

Now, assuming that all the work done is converted into heat energy we can replace symbol of 'work done' with that of 'heat produced'. So,

$$H = I^2 Rt$$

Applications of Heating Effect of Current

The heating effect of current is used in various electrical heating appliances such as electric bulb, electric iron, room heaters, geysers, electric fuse etc.

10.6ELECTRIC POWER

Electric power is the rate, per unit time, at which electric energy is transferred by an electric circuit.

Let P = Electric power

$P = W/t$

$P = V I$

Where, V is the applied voltage and I is the current flowing through the circuit. SI unit of power is Volt (V).

Now $P = V I$

If, V = 1 Volt (1 V) and I = 1 Ampere (1 A), then,

$P = 1$ Watt

Thus, power is said to be 1 watt, if a potential difference of 1 volt causes 1 ampere of current to flow through the circuit.

Bigger units of electric power are Kilo Watt (KW) and Mega Watt (MW)

10.7KIRCHHOFF'S LAWS

These two rules are commonly known as: Kirchhoff's circuit laws with one of Kirchhoff's laws dealing with the current flowing around a closed circuit, Kirchhoff's Current law (KCL) while the other law deals with the voltage sources present in a closed circuit, Kirchhoff's Voltage law, (KVL).

(i) Kirchhoff's First Law (Kirchhoff's Current Law) KCL

The law states that "The *algebraic sum of all the currents meeting at any junction point in an electric circuit is zero*"

$$\Sigma I = 0$$

Let us suppose the currents I_1, I_2, I_3 entering the junction are all positives in value and the two currents I_4, I_5 are leaving the junction are negative in values (Fig. 10.7), then according to KCL

$$I_1 + I_2 + I_3 - I_4 - I_5 = 0$$
$$I_1 + I_2 + I_3 = I_4 + I_5$$

Sum of incoming currents = sum of outgoing currents

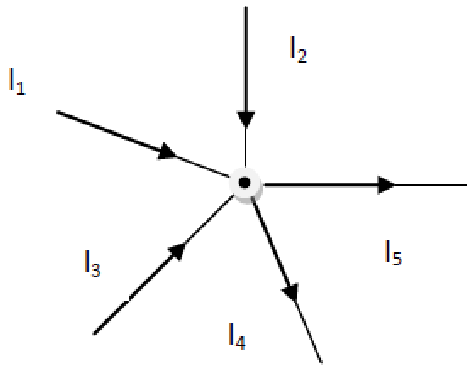


Figure 10.7

(ii) Kirchhoff's Second Law (Kirchhoff's Voltage Law) KVL

The law states that "*In any closed loop of a circuit, the algebraic sum of products of the resistances and currents plus the algebraic sum of all the e.m.f. in that circuit is zero*".

In any closed circuit; $\Sigma E + \Sigma IR = 0$

Here we use two sign conventions (Fig. 10.8).

1. If we go from negative terminal of the battery to the positive terminal then there is rise in potential and it is considered positive. And if we go from positive terminal to negative terminal, there is fall of potential and it is considered as negative.

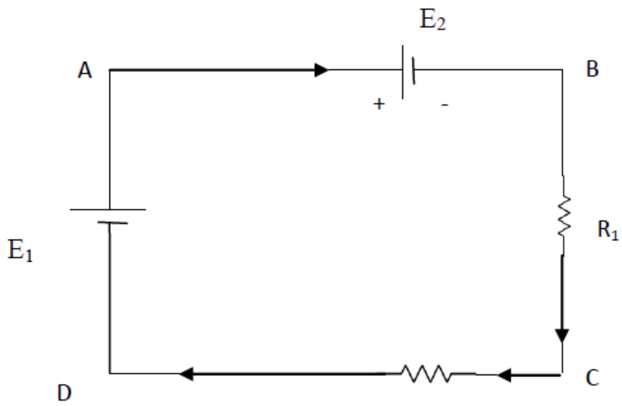


Figure 10.8

2. If we go with the current, voltage drop is negative and if we go against the current, the voltage drop is positive.

In the closed loop ABCD using KVL we get

$$- E_2 - IR_1 - IR_2 + E_1 = 0$$

Solved Numericals

Example 1. An emf source of 6V is connected to a resistive lamp and a current of 2 amperes flows. What is the resistance of lamp?

Solution. Given, $V = 6$ V and $I = 2$A

From Ohm's law, we know, $V = I R$ or $R = V/I$

$R = 6/2 = 3\Omega$

Example 2. An electric fan has a resistance of 100ohms. It is plugged into potential difference of 220 V. How much current passes through the fan?

Solution. Given, $R = 100$ ohms and $V = 220$ V

We know, $I = V/R = 220/100$

Therefore $I = 2.2$ A

Example 3. Calculate the total resistance if three resistances of 1 ohm, 2 ohm and 3 ohm are connected in series.

Solution. Given resistances, $R_1 = 1$ohm,

$R_2 = 2$ ohm

$R_3 = 3$ ohm

We know that in series combination; $R = R_1 + R_2 + R_3$

Therefore $R = 1 + 2 + 3 = 6$ ohm

Example 4. Calculate the total resistance if three resistances of 4 ohm, 1 ohm and 6 ohm are connected in parallel.

Solution. Given, $R_1 = 4$ohm

$R_2 = 1$ ohm

$R_3 = 6$ ohm

Form formula we know in parallel combination

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Hence $$\frac{1}{R} = \frac{1}{4} + \frac{1}{1} + \frac{1}{6} = \frac{3+12+2}{12} = \frac{17}{12}$$

Therefore, Total resistance, $R = \frac{12}{17}$ ohm

* * * * * *

EXERCISE

Multiple Choice Questions

1) The resistance of the wire varies inversely as:
   a) Area of cross section
   b) Resistivity
   c) Length
   d) Temperature
2) The curve representing Ohm's Law is a
   A. Linear
   B. Cosine Function
   C. Hyperbola
   D. Parabola
3) Resistance is a measure of a material's opposition to
   A. Voltage
   B. Current
   C. Electric Force
   D. Movement of protons
4) Give the name of materials which contain lots of free electrons
   A. Insulators
   B. Conductors
   C. Semi-conductors
   D. None of the above
5) Wire wound variable resistance is known as
   A. Capacitor
   B. Resistor
   C. Diode
   D. Rheostat
6) Specific resistance of a wire can be measured by formula
   A. R/L
   B. RA/L
   C. RL/A
   D. A/RL
7) Sum of all potential changes in a closed circuit is zero, stated law is called
   A. Kirchhoff's first rule
   B. Kirchhoff's second rule
   C. Kirchhoff's third rule
   D. Kirchhoff's fourth rule
8) Ohmic devices are devices that consequently
   A. Obey Ohm's law
   B. Obey Kirchhoff's law
   C. Violate Ohm's law
   D. Obey law of resistances.
9) Product of voltage and current is known as
   A. Work done
   B. Power
   C. Drift velocity
   D. E.M.F.

10) Current of 0.75 A, when a battery of 1.5 V is connected to wire of 5 m having cross sectional area 2.5×10^{-7} m², will have resistivity
   A. 1×10^{-7}
   B. 1.1×10^{-7}
   C. 2×10^{-7}
   D. 2.1×10^{-7}
11) SI unit of electric potential is
   A. Ampere
   B. Volt
   C. Joule
   D. Coulomb
12) SI unit of resistance is
   A. Ohm
   B. Henry
   C. Farad
   D. Newton
13) Conductance is reciprocal of
   A. Resistance
   B. Current
   C. Voltage
   D. Length
14) SI unit of specific resistance is
   A. Ohm
   B. Ohm metre
   C. Square of Ohm metre
   D. Farad

Short answer question

1. What is Electric current?
2. Define Resistance.
3. Define Specific resistance.
4. What is Conductance?
5. Explain Alternating current and Direct current.
6. Calculate the total resistance when resistances are connected in series.
7. Calculate the total resistance when resistances are connected in parallel.
8. Explain ohm's law.
9. Write short note on electric power.
10. Explain Kirchhoff laws.
11. If a wire is stretched to double of its length. What will be the new resistivity?

Long Answer type questions

1. Calculate the total resistance when resistances are connected in series and parallel.
2. Explain heating effect of current. Derive the formula for it and what are its applications?

3. a) Three resistors 1 Ω, 2 Ω and 3 Ω are combined in series. What is the total resistance of the combination?
   b) If the combination is connected to a battery of emf 12 V and negligible internal resistance, obtain the potential drop across each resistor.
4. Differentiate between AC and DC.
5. Explain Kirchhoff's law of current (KCL) and Kirchhoff's law of voltage (KVL).
6. If the resistance of a circuit is 12Ω and the current of 4 A passes through it calculate the potential difference. [Ans 48 V]
7. Electric fan takes a current of 0.5 amp when operated on a 200 V supply. Find the resistance. [Ans 440 ohms]
8. Calculate the total resistance when three resistances of 4 ohm, 8 ohm and 12 ohm are connected in series. [Ans 24 ohm]
9. Calculate the total resistance when resistances of 2 ohm and 2 ohm are connected in parallel. [Ans 1 ohm]
10. Calculate the power generated in a current of 2 A passes through a conductor having a potential difference of 220 V. [Ans 440 W]

Answers to multiple choice questions

1)	A	2)	A	3)	B	4)	B
5)	D	6)	B	7)	B	8)	A
9)	B	10)	A				

ELECTROMAGNETISM

Learning Objectives: After studying this chapter, students will be able to;

- *understand the magnetic field associated with flow of current and related parameters*
- *classify materials on basis of magnetic properties*
- *describe magnetic flux and magnetic lines of force*

11.1 ELECTROMAGNETISM

Electromagnetism or magnetism in general is the study of production of magnetic field when current is passed through a conductor. Various terms associated with magnetism are;

Magnetization (I)

It represents the extent to whicha material is magnetizedwhen placed in a magnetic field. It is given by magnetic moment per unit volume of material.

$$I = \frac{M}{V}$$

where, M is magnetic moment and V is volume of the material.

Unit: Ampere/meter

Magnetic Intensity (H):

It is the capability of magnetic field to magnetize a magnetic material.

Magnetic Permeability (μ):

It is property of material and defined as the degree to which magnetic lines of force can penetrate the medium.

Magnetic susceptibility (χ):

It is a property which determines how easily a specimen can be magnetised. It is given by ratio of magnetization and magnetic Intensity.

$$\chi = \frac{I}{H}$$

Types of Magnetic Materials:

On the basis of behaviour of magnetic material in magnetic field, the materials are divided in to three categories:

1. **Diamagnetic materials:**

The materials when placed in magnetic field, acquire magnetism opposite to the direction of magnetic field. The magnetic dipoles in these substances tend to align opposite to the applied field and tends to repel the external field around it.

- Diamagnetic substances have tendency to move from stronger to the weaker magnetic field.
- When rod of diamagnetic material is placed in magnetic field, it aligns perpendicular to the magnetic field.
- Permeability of diamagnetic material is < 1.

Examples; gold, water, mercury, graphite, lead etc

2. Paramagnetic materials:

Paramagnetic substances are those which get weakly magnetized when placed in an external magnetic field. These materials show weak attraction in magnetic field. The magnetic dipoles in the magnetic materials tend to align along the applied magnetic field. Such materials show weak feeble magnetization and the magnetization disappears as soon as the external field is removed.

- Permeability of parmagnetic material is > 1.
- The magnetization (**I**) of such materials dependent on the external magnetic field (**B**) and temperature (T) as:

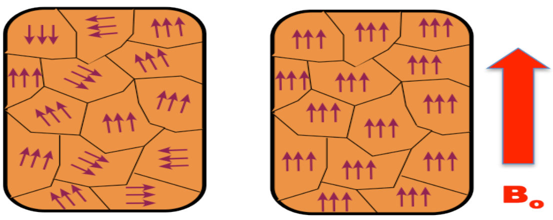
$$I = C\frac{B}{T}$$

Where C is the Curie constant.

Examples: sodium, platinum, liquid oxygen,salts of iron and nickel.

Ferromagnetic materials:

Ferromagnetic substances are those which get strongly magnetized when placed in an external magnetic field. They exhibit the strongest attraction in magnetic field. Magnetic dipoles in these materials are arranged into domains.



(a) (b)

Figure 11.1

These domains are usually randomly orientedas shown in Fig. 11.1 (a) and net magnetism is zero in the absence of magnetic field. When an external field is applied, the domains reorient themselves to reinforce the external field as shown in Fig. 11.1 (b) and produce a strong internal magnetic field that is along the external field.

These materials show magnetism on removal of magnetic field.

Examples are iron, cobalt, nickel, neodymium and their alloys. These are usually highly ferromagnetic and are used to make permanent magnets.

11.2 MAGNETIC FIELD

The space around a magnetic material or a moving electric charge within which the force of magnetism can be experienced.

Unit: Tesla (Wb/m^2)

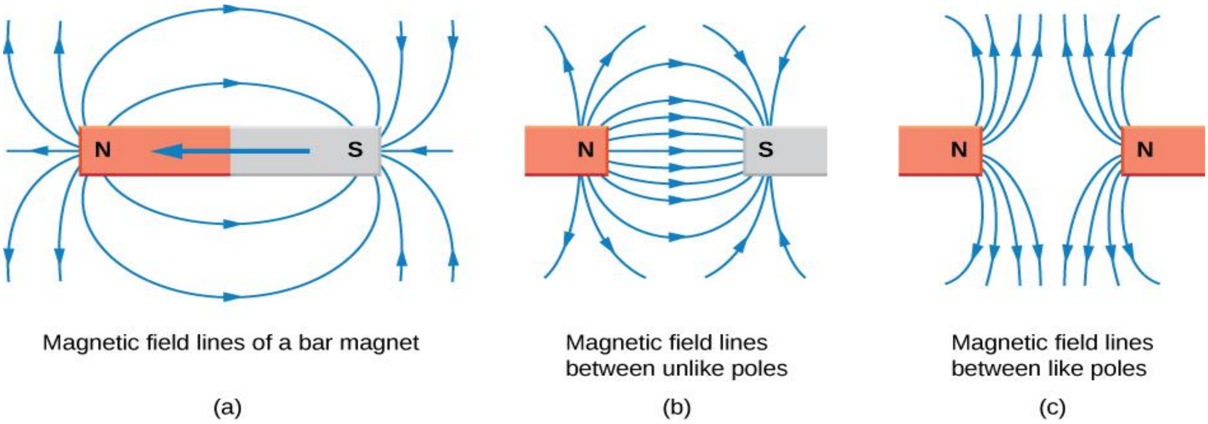


Figure 11.2

Magnetic lines of force:

Curved lines used to represent a magnetic field, drawn such that the number of lines relates to the magnetic field's strength at a given point(Fig. 11.2).

Properties of magnetic lines of force

(i) The magnetic field lines of a magnet form continuous closed loops.
(ii) The tangent to the field line at a given point represents the direction of the net magnetic field B at that point.
(iii) Larger the number of field lines crossing per unit area, the stronger is the magnitude of the magnetic field B.
(iv) Their density decreases with increasing distance from the poles.
(v) The magnetic field lines do not intersect with each other.
(vi) They flow from the South Pole to the North Pole within a material and North Pole to South Pole in air.

Magnetic flux:

The total number of magnetic field lines crossing through given surface area S held perpendicular to direction of magnetic field B.

$$\phi = B\ S\ Cos\theta$$

Unit: The SI unit of magnetic flux is the weber (Wb)

Magnetic Intensity:

Itis
the amount of magnetic flux in a unit area perpendicular to thedirection of magnetic flow.

11.3 ELECTROMAGNETIC INDUCTION

The induction of an electromotive force by the motion of a conductor across a magnetic field or by a change in magnetic flux in a magnetic field.is called **electromagnetic induction**.

It is used in electrical motor, generator etc. to generate electricity.

* * * * * *

EXERCISES

Multiple choice questions

1) The direction of a magnetic field within a magnet is
   A. Front to back
   B. north to south
   C. south to north
   D. west to south

2) When the speed at which a conductor is moved through a magnetic field is increased, the induced voltage
   A. Increases
   B. decreases
   C. reaches zero
   D. remains constant

3) An electromagnetic field exists only when there is
   A. Current
   B. voltage
   C. increasing current
   D. decreasing current

4) When the north poles of two bar magnets are brought close together, there will be
   A. No force
   B. Downward force
   C. Force of attraction
   D. Force of repulsion

5) E.M.F can be induced in a circuit by
   A. changing magnetic flux density
   B. changing area of circuit
   C. changing the angle
   D. all of above

6) Total number of magnetic field lines passing through an area is called
   A. magnetic flux density
   B. magnetic flux
   C. e.m.f
   D. voltage

7) Basic source of magnetism _______________.
   A. Charged particles alone
   B. Movement of charged particles
   C. Magnetic dipoles
   D. Magnetic domains

8) Example for para-magnetic materials
   A. super conductors
   B. alkali metals

C. transition metals
D. Ferrites

9) Example for ferro-magnetic materials
   A. super conductors
   B. alkali metals
   C. transition metals
   D. Ferrites

Short Answer type question

1. Define magnetic flux and write its unit.
2. Define electromagnetic induction with example
3. Define magnetic field.
4. What is unit of magnetic field?
5. What is magnetic susceptibility?
6. Write applications of EMI.
7. Define magnetic field intensity.
8. What is the relation between Magnetization and Magnetic field?

Long Answer Questions

1. What are magnetic lines of force? Write their properties.
2. Explain type of magnetic materials.
3. Explain ferromagnetic materials with their magnetic domains theory.
4. Explain difference between electric field and magnetic field.
5. Differentiate between paramagnetic and ferromagnetic materials with examples.
6. What is electromagnetic induction? Give its application along with its working.

Answers to multiple choice questions

1)	C	2)	D	3)	C	4)	D
5)	D	6)	B	7)	B	8)	A
9)	C						